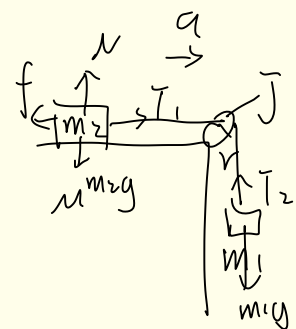


Chapter 3.

3-3.



解: (1) 对 m_2 : $T_1 - \mu m_2 g = m_2 a$ ①

对 m_1 : $m_1 g - T_2 = m_1 a$ ②

对滑轮: $T_2 r - T_1 r = J \cdot \beta$ ③

$\alpha = \beta \cdot r$ ④

由

$$\Rightarrow \begin{cases} a = \frac{m_1 g r^2 - \mu m_2 g r^2}{J + m_1 r^2 + m_2 r^2} \\ T_1 = \frac{m_1 m_2 g r^2 + J \mu m_2 g + \mu m_1 m_2 g r^2}{J + m_1 r^2 + m_2 r^2} \\ T_2 = \frac{J m_1 g + m_1 m_2 g r^2 + \mu m_1 m_2 g r^2}{J + m_1 r^2 + m_2 r^2} \end{cases}$$

(2) $T_1 = m_2 a$ ①

$m_1 g - T_2 = m_1 a$ ②

$T_2 r - T_1 r = J \cdot \beta$ ③

$\alpha = \beta \cdot r$ ④

$\Rightarrow$

$$a = \frac{m_1 g r^2}{J - m_1 r^2 + m_2 r^2}$$

$$T_2 = \frac{m_2 g J + m_1 m_2 g r^2}{J - m_1 r^2 + m_2 r^2}$$

$$T_1 = \frac{m_1 m_2 g r^2}{J - m_1 r^2 + m_2 r^2}$$

$$T_1 = m_2 a + \mu m_2 g$$

$$T_2 = m_1 g - m_1 a$$

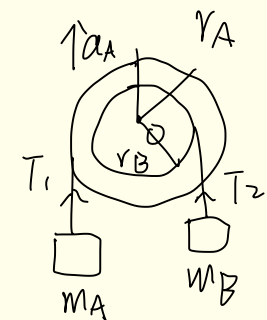
$$r^2 (m_1 g - m_1 a - m_2 a - \mu m_2 g) = J \cdot \frac{a}{r}$$

$$(m_1 g - \mu m_2 g) r^2 - (m_1 + m_2) a r^2 = J \cdot a$$

$$\frac{r^2 (m_1 g - \mu m_2 g) = a J = \frac{m_1 g r^2 - \mu m_2 g r^2}{J + m_1 r^2 + m_2 r^2}}$$

$$\begin{aligned} T_1 &= \frac{m_1 m_2 g r^2 - \mu m_2^2 g r^2}{J + m_1 r^2 + m_2 r^2} + \frac{\mu m_2 g (J + m_1 r^2 + m_2 r^2)}{J + m_1 r^2 + m_2 r^2} \\ &= \frac{m_1 m_2 g r^2 + J \mu m_2 g + \mu m_1 m_2 g r^2}{J + m_1 r^2 + m_2 r^2} \end{aligned}$$

3-4.



证: $m_B = \frac{J a_A + m_A r_A^2 (g + a_A)}{r_A r_B g - r_B^2 a_A}$

$m_A g - T_1 = m_A a_A$ ①

$m_B g - T_2 = m_B a_B$ ②

$T_2 r_A - T_1 r_B = J \cdot \beta$ ③

$$\Rightarrow m_B = \frac{J a_A + m_A r_A^2 (g + a_A)}{r_A r_B g - r_B^2 a_A}$$

$$J \quad a_A = r_A \beta \quad (4)$$

$$a_B = r_B \beta \quad (5)$$

3-12.

解: $J = 4 \times 10^3 \text{ kg} \cdot \text{m}^2$

$$n_1 = 30 \text{ r/min} = 0.5 \text{ r/s}, \quad n_2 = 10 \text{ r/min}$$

$$\omega_1 = \frac{2\pi n_1}{60} = \pi \text{ rad/s}$$

$$E_k = \frac{1}{2} J \omega_1^2 = \frac{1}{2} \times 4 \times 10^3 \times \pi^2 = 2 \times 10^3 \pi^2 (\text{kg} \cdot \text{m}^2/\text{s}^2)$$

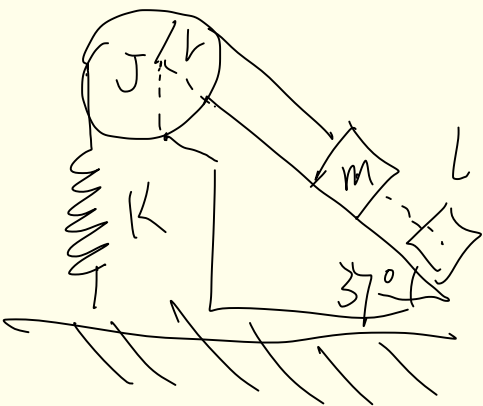
$$A = \frac{1}{2} J \omega_2^2 - \frac{1}{2} J \omega_1^2 \quad (1)$$

$$\omega_2 = \frac{2\pi n_2}{60} = \frac{\pi}{3} \text{ rad/s} \quad (2)$$

$$\Rightarrow A = 2 \times 10^3 \frac{8}{9} \pi^2 \quad (\text{kg} \cdot \text{m}^2/\text{s}^2)$$

$$\approx 1.77 \times 10^3 \pi^2$$

3-15.



$$J = 0.5 \text{ kg} \cdot \text{m}^2, \quad r = 30 \text{ cm}, \quad K = 20 \text{ N/m}, \quad m = 2 \text{ kg}$$

$$(1) \quad E_{k0} = 0, \quad E_{k1} = 0$$

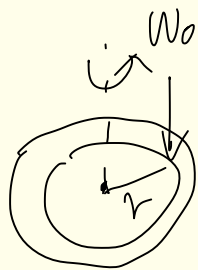
$$0 + 0 + 0 = 0 + \frac{1}{2} K \Delta L^2 - mg \Delta L \sin 37^\circ \quad (1)$$

$$\Rightarrow \Delta L = 1.2 \text{ m}$$

$$(2) \quad \frac{1}{2} K L^2 + \frac{1}{2} J \omega^2 + \frac{1}{2} m v^2 = mg L \sin 37^\circ \quad (2) \quad v = \frac{3\sqrt{17}}{5} \text{ m/s}$$

$$W = \frac{V}{r} \quad (5)$$

3-19.



1g/s.

$$J_1 = 5 \times 10^{-5} \text{ kg} \cdot \text{m}^2, \quad r = 0.1 \text{ m}.$$

$$\frac{1}{2} J \omega_0^2 = \frac{1}{2} J_1 \left(\frac{\omega_0}{2} \right)^2 \quad (1)$$

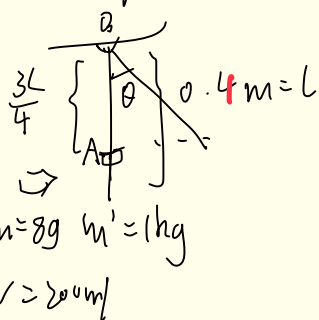
$$J_1 = (J + m r^2) \quad (2)$$

$$\Rightarrow t = 15 \text{ s}.$$

$$t = \frac{m}{v} \quad (3)$$

$$m \omega r L + m r^2 \omega.$$

3-21:



$$(1) J = m r^2 \quad (1) \quad v_i = \frac{3}{4} L \quad (2)$$

$$m v \cdot \frac{3}{4} L = m v \cdot \frac{3}{4} L + J \cdot \omega \quad (3) \Rightarrow \omega \approx 8.89 \text{ rad/s}$$

$$L - \frac{3}{4} L \cos \theta.$$

$$(2) \frac{1}{2} J \omega^2 + \frac{1}{2} m \left(\omega \frac{3}{4} L \right)^2 + m g \frac{L}{2} = 0 + 0 + m' g \left(L - \frac{L}{2} \cos \theta \right) + m g \left(L - \frac{3}{4} L \cos \theta \right) \quad (4)$$

$$+ m g \frac{L}{4} \Rightarrow \cos \theta \approx -0.076.$$

$$\theta = \arccos(-0.076) \approx 94^\circ.$$